Exact constant-field Landau–Lifshitz–Gilbert flow on the sphere

HCS Research Program

29 August 2026 (revision 1)

Abstract

We close the complete source-local dynamics of the constant-field Landau–Lifshitz–Gilbert equation on the unit sphere. In the stereographic coordinate $z = (m_1 + im_2)/(1 + m_3)$, the nonlinear vector field becomes $\dot{z} = (-\alpha\omega + i\omega)z$, while m_3 is an exact tanh trajectory. The energy law, north/south stability, pure-precession latitude periods, identity face, and every sampled-time fixed set follow in one boundary atlas. The periodic face is a continuum and therefore does not provide an isolated primitive-orbit or arithmetic determinant. Independent exact reconstruction and symbolic checks accompany the paper.

1 Model and exact chart

Let $e_3 = (0, 0, 1)$, $m(t) \in \mathbb{S}^2$, and $\alpha, \omega \geq 0$. We freeze

$$\dot{m} = -\omega m \times e_3 - \alpha\omega m \times (m \times e_3). \tag{1}$$

This is the normalized Landau–Lifshitz/Gilbert convention: the usual factor $1 + \alpha^2$ in the unreduced Gilbert equation has been absorbed into the declared frequency ω (equivalently, into the time unit). The vector field is tangent to the sphere. On $m_3 > -1$ put $z = (m_1 + im_2)/(1 + m_3)$. The inverse formulas are

$$m_1 + im_2 = \frac{2z}{1 + |z|^2}, \qquad m_3 = \frac{1 - |z|^2}{1 + |z|^2}. \tag{2}$$

Theorem 1 (Exact flow and pole atlas)**.** *For $-1 < m_3(0) < 1$, equation (1) has*

$$z(t) = z(0)e^{(-\alpha\omega+i\omega)t}, \qquad m_3(t) = \tanh(\alpha\omega t + \operatorname{artanh} m_3(0)). \tag{3}$$

The two poles are equilibria. With $E = 1 - m_3$,

$$\dot{E} = -\alpha\omega(1 - m_3^2) \leq 0. \tag{4}$$

If $\alpha\omega > 0$, the north pole is asymptotically stable and the south pole is unstable. Their transverse eigenvalues have real parts $-\alpha\omega$ and $+\alpha\omega$, respectively, and precession frequency ω.

Proof. Writing (1) in components gives $\dot{m}_1 = -\omega m_2 - \alpha\omega m_1 m_3$, $\dot{m}_2 = \omega m_1 - \alpha\omega m_2 m_3$, and $\dot{m}_3 = \alpha\omega(m_1^2 + m_2^2)$. The complex combination and the unit-sphere identity yield the scalar linear equation and the logistic equation in (3). Substitution in (2) proves reconstruction and (4). Linearizing the complex equation at the two chart ends gives the stated real parts; the pole cases themselves are immediate from (1). □

2 Periodic and sampled-time faces

Proposition 1 (All boundary cases)**.** *If $\alpha = 0$ and $\omega > 0$, latitude is conserved and every nonpolar point runs on a circle of period $2\pi/\omega$. If $\omega = 0$, the flow is the identity for every α. For a sampled time $\tau > 0$, when $\alpha\omega > 0$ the fixed set is exactly the two poles. On the pure precession face it is all of $\mathbb{S}^2$ iff $\omega\tau \in 2\pi\mathbb{Z}$, and otherwise it is the two poles; the $\omega = 0$ and $\tau = 0$ faces are identity maps.*

Proof. Equation (3) preserves $|z|$ exactly when $\alpha = 0$, and its phase is ωt. If $\alpha > 0$, the modulus can return to its initial value only at a pole. The listed resonance and identity alternatives then follow directly from the exponential flow. □

3 A complete source boundary ledger

The physical transverse radius is not the stereographic modulus: it is $r_\perp(t) = \sqrt{1 - m_3(t)^2} = 2|z(t)|/(1 + |z(t)|^2)$. Thus damping is exponential in the chart but nonlinear after sphere reconstruction. The following table records the exact fixed-set dimensions for representative faces; the evidence file contains the corresponding high-precision rows.

face	condition	flow type	sampled fixed set ($\tau > 0$)
identity	$\omega = 0$	constant	$\mathbb{S}^2$
precession	$\alpha = 0, \omega > 0$	latitude rotation	all or two poles
damped	$\alpha > 0, \omega > 0$	north attractor	two poles
north	$m = e_3$	equilibrium	north pole
south	$m = -e_3$	equilibrium	south pole

The pure-precession circles are a continuous family, not isolated primitive cycles. This distinction is essential: the exact ODE has a physical clock, but no canonical primitive-orbit product or target divisor.

References

[1] M. Lakshmanan, "The fascinating world of the Landau–Lifshitz–Gilbert equation: an overview," *Philosophical Transactions of the Royal Society A* 369(1935), 1280–1300 (2011). DOI.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; the source theorem contains no target arithmetic, Euler-factor, target-zero, automorphy or Hilbert–Pólya claim. **Data and code.** Exact formulas, regression rows and independent audit programs accompany HCS-C234. **AI-use disclosure.** Generative tools assisted drafting and code generation; internal checks validate the displayed identities. This is not external peer review.